$$\text{Then } \frac{dy}{dx} = (-1)(\cos x)^{-2}(-\sin x) \quad \text{(by chain rule)}$$

$$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x}.\frac{\sin x}{\cos x} = \sec x.\tan x \ .$$

That is, $\boxed{\frac{d}{dx}(\sec x) = \sec x \tan x}$.

(v) The cosecant function, cosec x

$$\text{Let } y = \operatorname{cosec} x = \frac{1}{\sin x} = (\sin x)^{-1} \ .$$

$$\frac{dy}{dx} = (-1)(\sin x)^{-2}(\cos x) \quad \text{(by chain rule)}$$

$$= -\frac{\cos x}{\sin^2 x} = -\frac{1}{\sin x}.\frac{\cos x}{\sin x} = -\operatorname{cosec} x \cot x \ .$$

That is, $\boxed{\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x}$.

(vi) The cotangent function, cot x

$$\text{Let } y = \cot x = \frac{\cos x}{\sin x} \ .$$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\cos x}{\sin x}\right)$$

$$= \frac{\sin x \frac{d}{dx}(\cos x) - \cos x \frac{d}{dx}(\sin x)}{\sin^2 x}$$

$$= \frac{\sin x(-\sin x) - \cos x(\cos x)}{\sin^2 x}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\frac{1}{\sin^2 x} = -\operatorname{cosec}^2 x$$

That is, $\boxed{\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x}$.

(6) The derivatives of the inverse trigonometric functions

(i) The derivative of arcsin x or $\sin^{-1} x$

$$\text{Let } y = f(x) = \sin^{-1} x \ .$$

$$\text{Then } y + \Delta y = f(x + \Delta x) = \sin^{-1}(x + \Delta x)$$

This implies, $x = \sin y$ and

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